

Atomically clean free-standing two-dimensional materials through heating in ultra-high vacuum

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Surface contamination not only influences but in some cases even dominates the measured properties of two-dimensional (2D) materials. Although different cleaning methods are often used for contamination removal, commonly used spectroscopic cleanliness assessment methods can leave the level of achieved cleanliness ambiguous. Therefore, despite two decades of research on 2D materials, the true cleanliness of the used samples is often left open to interpretation. While cleaning procedures for graphene have been extensively employed and studied[1], there is a lack of reporting on the cleaning of monolayer hexagonal boron nitride, particularly at the atomic level, despite its various projected applications, such as being a dielectric for nanoelectronic devices, and for quantum information and sensor technology due to its quantum light emission across a wide spectral range. In this work, free-standing monolayer graphene and hexagonal boron nitride are annealed at different temperatures in a custom-built ultra-high vacuum heating chamber, connected to a scanning transmission electron microscope via a vacuum transfer line[2], enabling atomically resolved cleanliness characterization as a function of annealing temperature, while eliminating the introduction of airborne contamination during sample transport. We find that, while annealing at 200°C already reduces contamination significantly, it is not until 400°C or higher, where over 90% of the free-standing monolayer areas are atomically clean. At this point, further contamination removal is mainly limited by defects in the material and metal contamination introduced during the sample transfer or growth. In addition, we find annealing-induced defects to be negligible in quantity, which provides the pristine, atomically clean areas necessary for further structural manipulations, such as defect or pore creation through more controlled techniques like electron or ion irradiation[3,4].

[1] Zhuang, B. et al., *Carbon* **173**, 609-636 (2021).

[2] Mangler, C. et al., *Microscopy and Microanalysis* **28**, 2940-2942 (2022).

[3] Trentino, A. et al., *Nano Letters* **21**, 5179-5185 (2021).

[4] Byrne, D.O. and Allen, F.I., *ACS Applied Nano Materials* **8**, 4565-4572 (2025).